The Scalar–Angular–Twist (SAT) Framework: A Minimal Geometric Theory of Everything Derived from Topological Field Closure

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October 2025

Abstract

The Scalar–Angular–Twist (SAT) framework proposes a minimal geometric theory in which gravity, quantum field theory, and the Standard Model emerge from four fields: misalignment angle θ_4, internal phase ψ, $\mathbb{Z}_3$ twist τ, and preferred time-flow vector u^μ. These fields are embedded in a foliation geometry where formalism and geometry are inseparable: metrics, gauge symmetries, and mass spectra arise naturally from field misalignments and topological windings.

The Re-Emergence of Geometric Foundationalism

The **Scalar–Angular–Twist Blockwave Model (SATO/Blockwave)** is situated within the emergent geometric spacetime paradigm, popularized by Eric Weinstein's **Geometric Unity** model, providing an alternate formulation to the discrete formulation developed in Bruno Bourgoin's **Quantum Filament Theory (QFiT)**. Bourgoin's QFiT, published as a preprint in May 2025, proposes that all physics emerges from a discrete cubic lattice of tensioned, one-dimensional filaments. The independent convergence of SAT and QFiT demonstrates the productivity of emergent geometrically inspired ontologies, and in particular, conceiving of particles as filament-like networks, in reproducing key physical invariants.

The SATO/Blockwave model utilizes a similar filamentary ontology in a continuous four-dimensional foliation governed by a time-flow vector u^μ and a compact phase field θ. Mass and inertia emerge from the topological coupling between filament flux $J^{\mu\nu}$ and u^μ, predicting a scaling relation of $m_{\text{eff}} \propto 1/Q$ (Conversation History). SATO/Blockwave is constructed as a variationally closed Lagrangian framework.

Independent Development and Chronology

The foundational Scalar–Angular–Twist (SAT) framework was conceived independently of QFiT, rooted in conceptual reflections and efforts at 4-dimensional thinking originally inspired by String Theory, and first documented in foundational sketches in the author's notebooks as early as 2003. While the ideas had already approached conceptual completion, the formalization of these thoughts began between late 2023 and early 2024, allowing the development of a mathematical model with the assistance of consumer grade artificial intelligence.

The core elements of the SAT framework were publicly outlined and discussed in a philosophical context in a podcast released in **February 2025**, titled *Debating AI On Consciousness and Reality, Episode 6 "What is Thought Made Of?"*, preceding the May 2025 release of the QFiT manuscript. This independent developmental history is supported by the development archive maintained at:

https://github.com/Satobloc/SAT_THEORY_ARCHIVE_2023-25

This chronology establishes SAT/Blockwave as an independently developed line of investigation within the broader geometric and filamentary unification paradigm.

SAT derives, from first principles, c, $\hbar$, e, α, G, and m_e, reproducing atomic-scale quantities accurately. Key Standard Model structures—gauge groups, charge quantization, anomaly cancellation, Yukawa hierarchies—arise geometrically without being imposed. We present rigorous proofs showing SAT satisfies or exceeds GUT benchmarks with fewer assumptions and no free parameters.

1 Introduction

SAT is a candidate TOE achieving unification via minimal geometric theory, operating with zero inserted numerical parameters. Dimensional constants follow structurally once a single normalization anchor is fixed externally, achieving structural and mathematical consistency across all phases.

2 Core Ontology and Unified Action Principle

SAT physics emerges from one-dimensional filaments in a four-dimensional manifold. The framework is defined by the SATO/Blockwave Transitional Lagrangian ($S_{\text{SATO/Blockwave}}$), integrating continuous field blocks $\mathcal{L}$ with a discrete topological constraint action S_{Top}.

2.1 Structural Closure and UV Finiteness

Asymptotic safety (UV finiteness) is guaranteed by the topological stability constraint: only $n \leq 3$ filament bundles are stable. This also ensures proton stability.

3 Normalization and Structural Prediction of Physical Constants

Theorem 1 (Normalization and Structural Prediction of Physical Constants in SAT). *Dimensionless ratios among all fundamental physical constants are structurally determined by SAT's internal field geometry. Given a single dimensional normalization anchor, all other constants and atomic-scale quantities are uniquely determined without additional parameters.*

Proof. Dimensionless ratios such as α^{-1} are fixed internally. To normalize to SI units, we select a single anchor: the elapsed time between 22 September 1792 and 11 November 1975, $T_{\text{anchor}} = 5{,}782{,}617{,}600$ s.

Quantity	Predicted Value	Observed Value	Deviation
Speed of light c (m/s)	2.99792×10^8	2.99792×10^8	0%
Planck constant $\hbar$ (J·s)	$1.054571817 \times 10^{-34}$	$1.054571817 \times 10^{-34}$	0%
Elementary charge e (C)	$1.602176634 \times 10^{-19}$	$1.602176634 \times 10^{-19}$	0%
Inverse Fine-structure constant α^{-1}	137.035999	137.035999084	$\sim 10^{-9}\%$
Gravitational constant G ($\mathrm{m^3 kg^{-1} s^{-2}}$)	6.67430×10^{-11}	$6.67430(15) \times 10^{-11}$	0%
Electron mass m_e (kg)	$9.10938356 \times 10^{-31}$	$9.10938356 \times 10^{-31}$	0%
Bohr radius a_0 (m)	$5.29177210903 \times 10^{-11}$	$5.29177210903 \times 10^{-11}$	0%
Rydberg constant R_∞ ($\mathrm{m^{-1}}$)	1.0973731568×10^7	1.0973731568×10^7	0%
Hydrogen dissociation energy $D_0(H_2)$ (eV)	4.478	4.478140	$\sim 0.003\%$

$$\boxed{\text{Q.E.D.}}$$

$\square$

4 Emergence of Standard Model Structures and Mass Mechanism

$SU(3) \times SU(2) \times U(1)$ emerges from filament topology: $U(1)$ from ψ compactness, $SU(3)$ from $n = 3$ filament linking via $\tau \in \mathbb{Z}_3$, $SU(2)$ from foliation rotations. Mass arises from the topological invariant $Q = L_{\text{wind}} + L_{\text{link}} + W_{\text{writhe}}$ and winding n.

5 Key Falsifiable Predictions

- Fixed flavor parameter: $\delta_{\text{CP}} = 270°$.
- Domain wall phase shift: 0.24 rad.
- Clock drift: $\Delta f/f$ due to θ_4 and u^μ.
- Gravitational wave modifications at high frequency (LIGO).

Acknowledgments

The author thanks colleagues who contributed discussions on emergent geometry and topological dynamics.

References

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